

FOREIGN KEY constraint:

- It is also called as REFERENTIAL INTEGRITY CONSTRAINT.
- It designates a column as Foreign Key and establishes a relation between the Foreign Key and a specified Primary or Unique Key.
- A Composite Foreign Key designates a combination of columns as the Foreign Key.
- The Table or View containing the Foreign Key is called the child Object.
- The Table or View containing the REFERENCED KEY is called the parent object.
- The Foreign key and the Referenced Key can be in the same Table or View.
- The corresponding column or columns of the foreign key and the Referenced Key must match in order and data type.
- A Foreign key Constraint can be defined on a single key column either inline or out of line.
- A Composite Foreign Key on attributes should be declared at table level or out of line style.
- A Composite Foreign Key Constraint must refer to a Composite Unique Key or a Composite Primary Key in the Parent Table or View.
- The same column or combination of columns can be designated as both a Foreign Key and a Primary or Unique Key.
- We can designate the same column or combination of columns as both a foreign key and a Primary or Unique Key.

```
SQL> create table dept1(dno number(5) primary key, dname varchar2(20), address
varchar2(20));
```

Table created.

```
SQL> desc dept1
```

Name	Null?	Type
DNO	NOT NULL	NUMBER(5)
DNAME		VARCHAR2(20)
ADDRESS		VARCHAR2(20)

```
SQL> insert into dept1 values(&dno,&dname,&address);
```

Enter value for dno: 10

Enter value for dname: 'it'

Enter value for address: 'narasaraopet'

old 1: insert into dept1 values(&dno,&dname,&address)

new 1: insert into dept1 values(10,'it','narasaraopet')

1 row created.

```
SQL> /
```

Enter value for dno: 11

Enter value for dname: 'cse'

Enter value for address: 'guntur'

old 1: insert into dept1 values(&dno,&dname,&address)

new 1: insert into dept1 values(11,'cse','guntur')

1 row created.

SQL> /

Enter value for dno: 12

Enter value for dname: 'ee'

Enter value for address: 'guntur'

old 1: insert into dept1 values(&dno,&dname,&address)

new 1: insert into dept1 values(12,'ee','guntur')

1 row created.

SQL> select * from dept1;

DNO	DNAME	ADDRESS
10	it	narasaraopet
11	cse	guntur
12	ee	guntur

SQL> create table student1(sno number(5),sname varchar2(20),dno number(5),foreign key(dno) references dept1(dno));

Table created.

SQL> desc student1;

Name	Null?	Type
SNO		NUMBER(5)
SNAME		VARCHAR2(20)
DNO		NUMBER(5)

SQL> insert into student1 values(&sno,&sname,&dno);

Enter value for sno: 123

Enter value for sname: 'john'

Enter value for dno: 11

old 1: insert into student1 values(&sno,&sname,&dno)

new 1: insert into student1 values(123,'john',11)

1 row created.

SQL> /

Enter value for sno: 234

Enter value for sname: 'jack'

Enter value for dno: 44

old 1: insert into student1 values(&sno,&sname,&dno)

new 1: insert into student1 values(234,'jack',44)

insert into student1 values(234,'jack',44)

*

ERROR at line 1:

ORA-02291: integrity constraint (SCOTT.SYS_C00733) violated - parent key not found

```
SQL> delete from dept1 where dno=11;
delete from dept1 where dno=11
      *
```

ERROR at line 1:

ORA-02292: integrity constraint (SCOTT.SYS_C00733) violated - child record found.

On delete Cascade: The default behavior of the foreign key can be changed by using ON DELETE CASCADE option. When ON DELETE CASCADE option is specified in the foreign key definition, if the user deletes a record in the master table, all corresponding records in the detail table (child table) along with the record in the master table will be deleted.

Syntax:

Create table <table name> (Columnname datatype[size], Columnname datatype[size], Foreign key(column name) References<table name>(column name) ON DELETE CASACADE));

Example:

Create table student (Sno number(5), Sname varchar2(20), dno number(5), foreign key(dno) references dept(dno) ON DELETE CASCADE;
Table Created.

```
SQL>desc student
```

Name	Null?	Type
Sno		Number(5)
Sname		Varchar2(20)
Dno		Number(5)

```
SQL>Delete from student;
```

1 row deleted.

```
SQL>Delete from dept;
```

4 rows deleted.